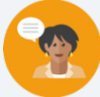


13.1 Parallel and Perpendicular Lines

Lesson Introduction

 Benchmark	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.		 Learning Target	I can use coordinate geometry to classify parallel and perpendicular lines.
 Benchmark Coherence	Building On MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.		Working Toward N/A	
 Essential Questions	- What characteristics do slopes of parallel lines have? - What characteristics do slopes of perpendicular lines have? - How can a linear equation be converted from standard form to slope-intercept form? - How can the slope be identified in slope-intercept form?			
 Lesson Narrative	The lesson guides students to discover that parallel lines have the same slope, and that the slopes of perpendicular lines are opposite reciprocals. Students work with graphs and equations to identify the slopes. The lesson also requires students to convert from the standard form of a linear equation to the slope-intercept form of an equation.			
 Component Summary	13.1.1 Warm-Up (~5 Min)	Artificial Intelligence and Algorithms (<i>SE p. 281</i>)	13.1.3 Guided Instruction (15 Min)	Formal Introduction of Parallel and Perpendicular Lines (<i>SE p. 284</i>)
	13.1.2 Collaboration (15 Min)	Deriving the Relationships for Parallel and Perpendicular Lines (<i>SE p. 282</i>)	13.1.4 Guided Practice (10 Min)	Practice on Identifying Parallel and Perpendicular Lines (<i>SE p. 286</i>)
	13.1.5 Wrap-Up (~5 Min)	Students summarize their learning (<i>SE p. 287</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Parallel - Perpendicular <u>Additional Terms:</u> - Slope		- Career Profile Video (optional) Graphing Utility	13.1.1 Warm-Up contains a video to accompany the question prompts. - If choosing to make 13.1.2 question 13 into a Desmos Card Sort, create the Card Sort.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may make computational errors when converting from standard form to slope-intercept form. - Students may identify lines as perpendicular if the slopes are opposite, instead of opposite reciprocals.	Consider using a graphing utility, such as Desmos or GeoGebra, with this lesson.
	Literacy and Language Routines Think-Pair-Share There are multiple opportunities highlighted throughout the lesson to Think-Pair-Share. Students work individually and then collaborate. Give students the time to think individually before sharing with a partner or group in the multiple exercises during the span of the lesson.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Use Patterns and Structure Lesson 13.1 requires students to apply patterns and structures to derive the relationships between parallel and perpendicular lines. Throughout the practice exercises in the lesson, students should look for similarities of characteristics of parallel and perpendicular lines.
 Differentiate Instruction	Lesson 13.1 lends itself to differentiate to a visual approach. Consider either using an online graphing utility with the class, or having students use a graphing calculator on personal devices. The Collaboration involves a large amount of graphing and writing equations. Desmos can also highlight the relationship between standard form of a linear equation and the slope-intercept form of a linear equation.	
Lesson Notes		

13.1.1 Warm-Up: Artificial Intelligence and Machine Learning

Component Overview: Students read a short narrative and watch a video about artificial intelligence and algorithms. Then, they are asked to respond to three questions.



13.1.1 Warm-Up: Artificial Intelligence and Machine Learning

Artificial intelligence enables a machine to simulate human behavior and decisions. Machine learning is when a machine uses artificial intelligence to learn from past data without a programmer having to re-program the machine.

Many of the women in the video used the word “algorithm.”

1. Describe one task you would like a machine to be programmed to do.

Answers will vary. Drive my car.

Both artificial intelligence and machine learning rely on mistakes to help improve the programming.

2. Describe a mistake you have made that helped you improve.

Answers will vary. Making mistakes helps me to see how a problem works correctly and then I can understand and remember the procedures for the problem.



Consider asking the following questions to guide student thinking and responses:

- What are some forms of artificial intelligence that are used in everyday life?
- How are algorithms used in the study of mathematics?

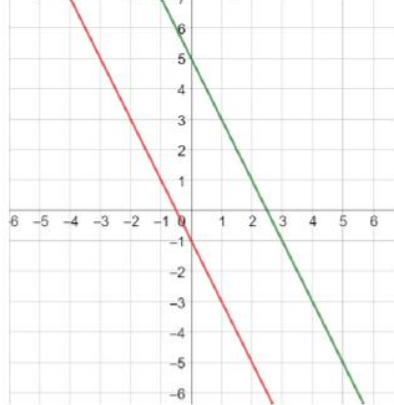
13.1.2 Collaboration: Parallel and Perpendicular Lines

Component Overview: This Collaboration has students apply multiple skills for discovering the characteristics of parallel lines, perpendicular lines, and their slopes. Students will be required to graph by hand and convert from the standard form of a linear equation to the slope-intercept form of an equation.



13.1.2 Collaboration: Parallel and Perpendicular Lines

1. Graph the equation $y = -2x + 5$ on the coordinate plane.



2. Rewrite the equation $6x + 3y = -3$ in slope-intercept form.
 $y = -2x - 1$
3. Graph the equation $6x + 3y = -3$ using slope-intercept form.
See above
4. What similarities do the two equations have?
They have the same slope.
5. What differences do the two equations have?
They have different y-intercepts.



Students should be collaborating throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions, and which students help clarify.



Consider having students draw slope triangles or count the slope on the graph to provide a visual approach.



Monitor student work when converting from standard form to slope-intercept form. Students may forget to subtract $6x$ from each side, and may forget to divide each term by 3 to isolate y .

13.1.2 Collaboration: Parallel and Perpendicular Lines (continued)

6. Complete the statement.

The equations $y = -2x + 5$ and $6x + 3y = -3$ resulted in

- ☒ parallel
- ☐ perpendicular
- ☐ vertical

lines.

The slopes of the equations are

- ☒ equal.
- ☐ undefined.
- ☐ opposite reciprocals.

7. Rewrite the equation $4y - 6x = 8$ in slope-intercept form.

$$y = \frac{3}{2}x + 2$$

8. Graph the equation $4y - 6x = 8$ using slope-intercept form.



9. Graph the equation $y = -\frac{2}{3}x - 4$.

See above.

10. What do you notice about the slopes in the two equations?

They are the opposite reciprocal of each other.



For a kinesthetic approach to questions 8-9, consider having students work in partners. Have one student graph the equation on question 8, and have another student graph the equation on question 9.



Think-Pair-Share

Provide students with the opportunity to share with a partner what the students notice about the graph and the slopes.

13.1.2 Collaboration: Parallel and Perpendicular Lines (continued)

11. Complete the statement.

Equation $4y - 6x = 8$ and equation $y = -\frac{2}{3}x - 4$ form

- ☐ parallel
- ☒ perpendicular
- ☐ vertical

lines; therefore, the slopes of the

equations are

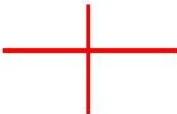
- ☐ equal.
- ☐ undefined.
- ☒ opposite reciprocals.



Think-Pair-Share

Have students complete questions 11 and 12 as a Think-Pair-Share. Give students a few seconds to ponder the two questions, and then have students share their responses.

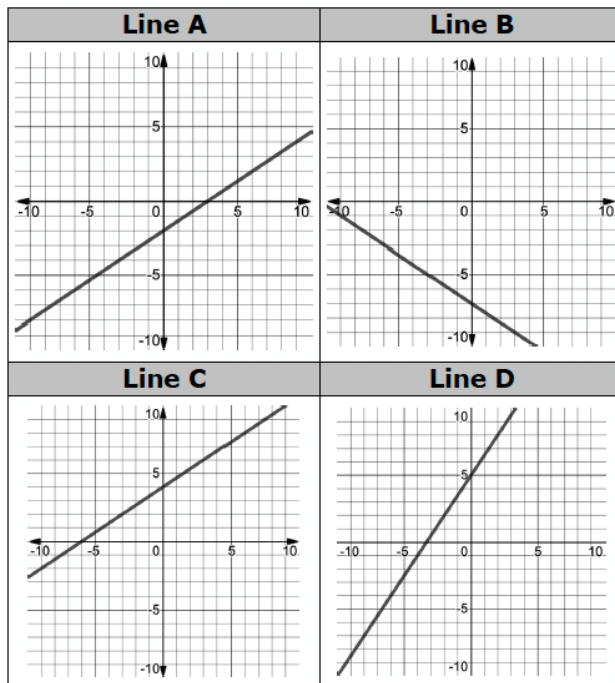
12. Complete the table.

Type	Sketch	Slope
Parallel lines		<i>Equal</i>
Perpendicular lines		<i>Opposite Reciprocals</i>

13.1.2 Collaboration: Parallel and Perpendicular Lines (continued)

13. Match the two lines that are parallel and the two lines that are perpendicular. Then, match the equations.

	Lines	Equations
Parallel	A & C	E & G
Perpendicular	B & D	F & H



Equation E	Equation F
$y + 2 = \frac{2}{3}(x + 9)$	$2x + 3y = -21$
Equation G	Equation H
$-2x + 3y = -6$	$y = \frac{3}{2}x + 5$



For a challenge, consider having students write the equations of each line by only giving them the graphs.



Consider making question 13 into a Desmos Card Sort for a digital extension.



For Lines A-D, ask students about the strategies they will choose to sort the lines.

For Equations E-H, ask students to share their strategies for matching the graphs to the equations.

13.1.3 Guided Instruction: Parallel and Perpendicular Lines

Component Overview: Students work through two questions on parallel and perpendicular lines.



13.1.3 Guided Instruction: Parallel and Perpendicular Lines



Lines are **parallel** if the slopes of the lines are the same.

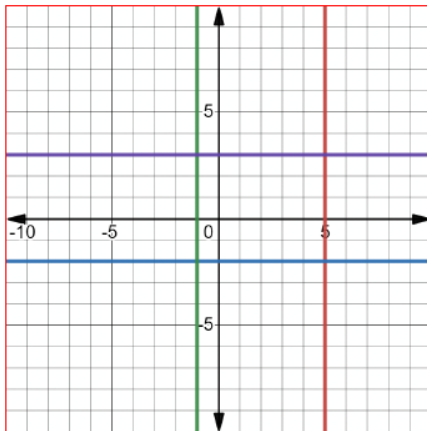


Lines are **perpendicular** if the slopes are opposites and reciprocals.



This section of the lesson provides students the opportunity to refine and apply what was learned in the Guided Discovery. Question 1 gives students a graph and asks students to apply properties with a rectangle. On question 2, students are given a graph and must find whether two lines are parallel. Allow students the opportunity to discuss what they notice about the four lines in question 1.

1. Four lines are shown on the coordinate plane.



- a. What do you notice about the four lines?
Answers will vary. The purple and blue lines are parallel. The green and red lines are parallel. The purple line is perpendicular to the green and red lines, the blue line is perpendicular to the green and red lines.

13.1.3 Guided Instruction: Parallel and Perpendicular Lines (continued)

- b. In the graph of the four lines, the four lines create a rectangle. Explain how each of the properties of a rectangle could be shown.

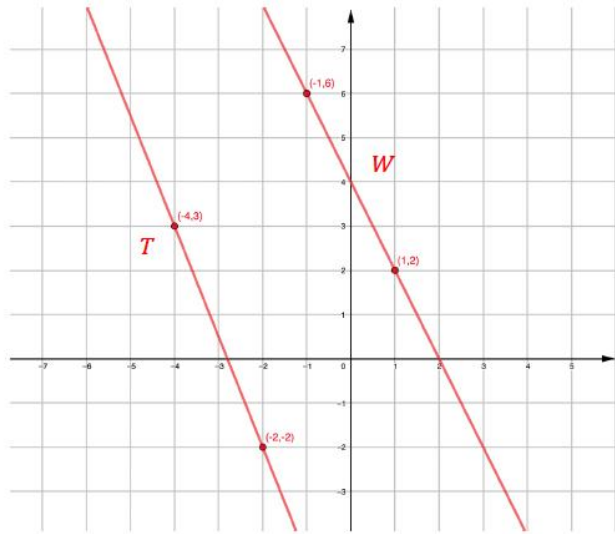
Property	Explanation
Opposite sides are parallel.	<i>The purple and blue lines are opposite each other and parallel because they have the same slope. The green and red lines are opposite each other and parallel because they have the same slope.</i>
All angles are right angles.	<i>The purple and blue lines are perpendicular (opposite reciprocal slopes) to the red line forming eight right angles. The purple and blue lines are perpendicular (opposite reciprocal slopes) to the green line forming eight right angles.</i>



Consider making question 1b a Think-Pair-Share.

13.1.3 Guided Instruction: Parallel and Perpendicular Lines (continued)

2. Line W has points $(-1, 6)$ and $(1, 2)$ on it. Line T has points $(-4, 3)$ and $(-2, -2)$ on it.



- Graph line W and line T .
- Determine if line W and line T are parallel lines, perpendicular lines, or neither. Justify your answer with math.

The slope of $W = \frac{2 - 6}{1 - (-1)} = -2$

The slope of $T = \frac{-2 - 3}{-2 - (-4)} = \frac{-5}{2}$

The lines are neither parallel nor perpendicular because their slopes are neither the same nor opposite reciprocals.



Consider having students draw slope triangles on the graph.



For a challenge, have students create and graph a line that is parallel to either line W or line T . For an additional extension, have students graph a line that is perpendicular to either line W or line T .

13.1.3 Guided Instruction: Parallel and Perpendicular Lines (continued)

3. Line N is written as $y = \frac{1}{3}x - 8$. Line V is written as $6x + 2y = 4$.

Determine if line N and line V are parallel lines, perpendicular lines, or neither. Justify your answer with math.

$$y = \frac{1}{3}x - 8$$

$$6x + 2y = 4$$

$$2y = -6x + 4$$

$$y = -3x + 2$$

The lines are perpendicular because their slopes are opposite reciprocals.



Monitor students' work on question 3, as there are multiple errors that can be made when converting from standard form to slope-intercept form.

13.1.4 Guided Practice: Parallel and Perpendicular Lines

Component Overview: Students work with a partner or group on two separate tasks of determining parallel and perpendicular lines.



13.1.4 Guided Practice: Parallel and Perpendicular Lines

1. Given that line M is $3x - y = 24$, determine if the following lines are parallel to line M .

Complete the table.

Equation of Line	Work	Is the line parallel to line M ?
$y = 3x + 8$	$y = 3x - 24$, slope = 3 $y = 3x + 8$, slope = 3	☐ Yes ☐ No
$y = -3x - 1$	$y = 3x - 24$, slope = 3 $y = -3x - 1$, slope = -3	☐ Yes ☐ No
$3y = x - 4$	$y = 3x - 24$, slope = 3 $y = \frac{1}{3}x - \frac{4}{3}$, slope = $\frac{1}{3}$	☐ Yes ☐ No
$3x - y = 3$	$y = 3x - 24$, slope = 3 $y = 3x - 3$, slope = 3	☐ Yes ☐ No



Consider asking the following question to students as they begin 13.1.4.



After converting $x - y = 24$ from standard form to slope-intercept form, what strategies can you use to determine whether the lines are parallel to line M ?

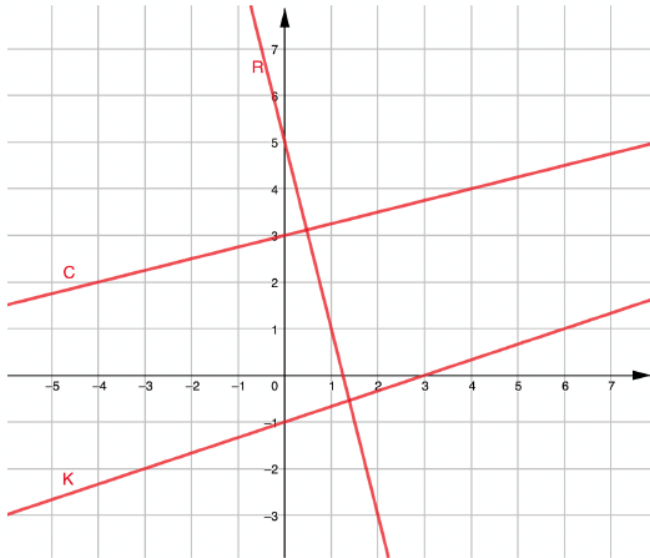
Look for students who may have a common error of forgetting to divide by -1 after subtracting x from each side when converting $x - y = 24$ from standard form to slope-intercept form.



For students who finish early, challenge them to create a linear function that would be parallel and perpendicular to $x - y = 24$

13.1.4 Guided Practice: Parallel and Perpendicular Lines (continued)

2. Line R is written as $4x + y = 5$, line K is written as $12y = 4x - 12$, and line C is written as $4y - 12 = x$.



- Graph lines R , K , and C .
Images drawn on graph.
- Determine which two lines are perpendicular lines.
Justify your answer with math.

$$R; y = -4x + 5, \text{ slope} = -4$$

$$K; y = \frac{1}{3}x - 1, \text{ slope} = \frac{1}{3}$$

$$C; y = \frac{1}{4}x + 3, \text{ slope} = \frac{1}{4}$$

Lines R and C are perpendicular to each other because their slopes are opposite reciprocals.



Consider asking students the following question before beginning question 2:
Which lines do you think will be perpendicular to one another? Explain your reasoning.



Consider having students do a Think-Pair-Share for questions 2a and 2b. Have students work silently, then pair with another classmate to share.

13.1.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



13.1.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.

Answers will vary.